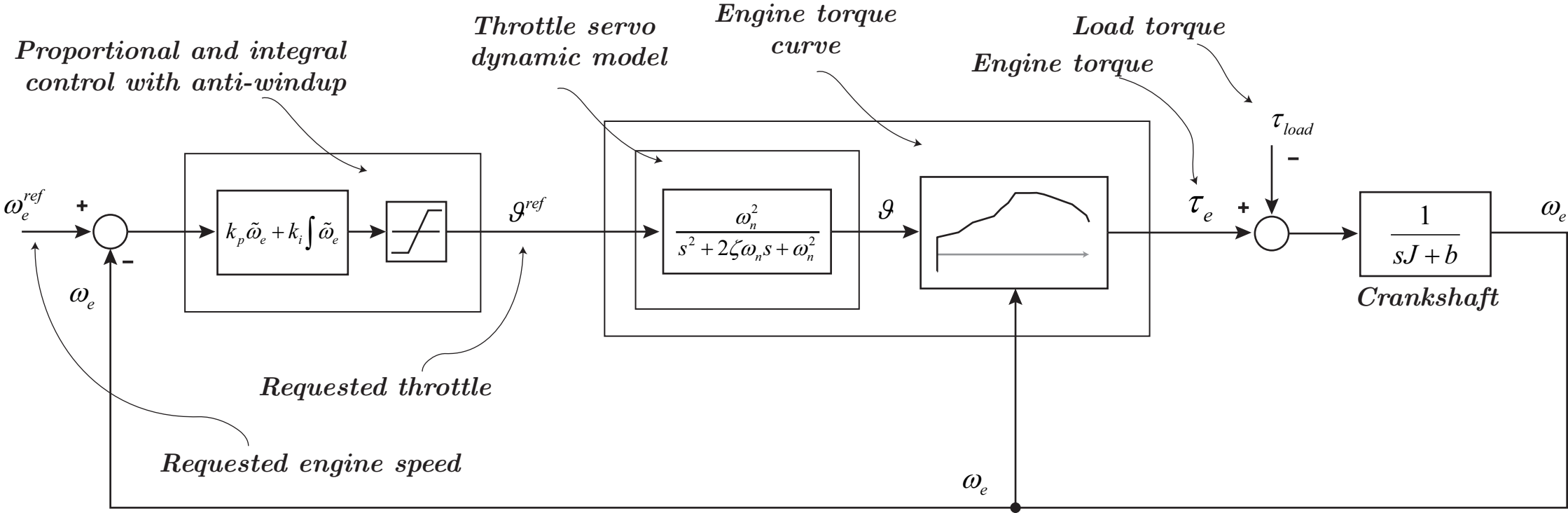
Proportional and integral control with anti-windup
Throttle servo dynamic model
Engine torque curve
Load torque
Engine torque
$\omega_e^{ref}$
+
−
$\omega_e$
$k_p\tilde{\omega}_e + k_i\int\tilde{\omega}_e$
$\vartheta^{ref}$
$\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$
$\vartheta$
$\tau_e$
$\tau_{load}$
$\frac{1}{sJ+b}$
Crankshaft
$\omega_e$
Requested throttle
Requested engine speed
$\omega_e$